IX WORKSHOP BRASILEIRO DE MICROMETEOROLOGIA
11-13 Novembro de 2015
Santa Maria, RS - Brasil

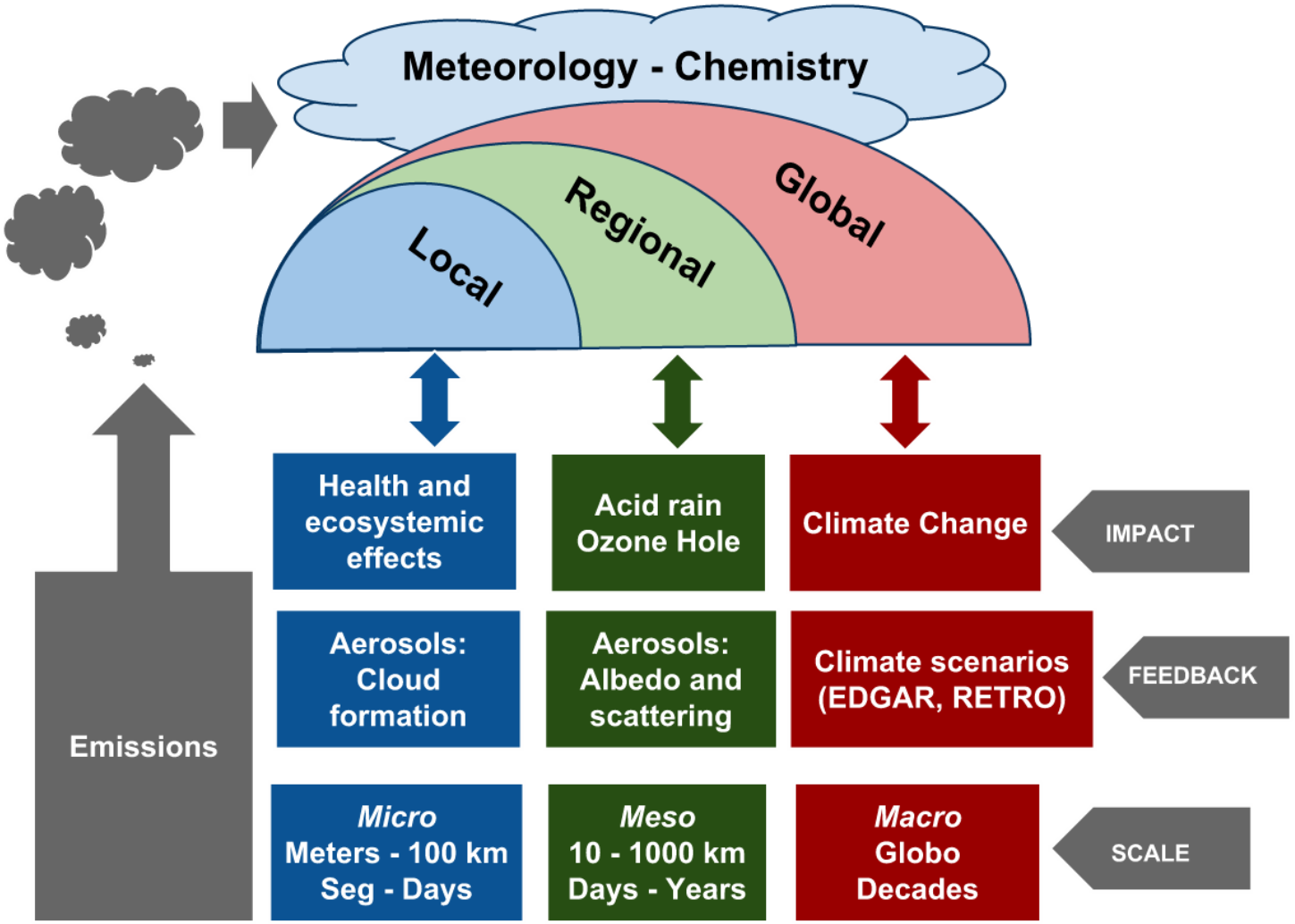
High resolution air pollutant simulation for the Metropolitan Region of Porto Alegre

Sergio Ibarra, Angel Vela, Amanda Rehbein, Rita Yuri

Institute of Astronomy, Geophysics and Atmospheric Sciences
University of São Paulo

sergio.ibarra@iag.usp.br

Motivation



Scheme of impacts of emissions in biosphere (Ibarra, 2014)

Motivation

Quim. Nova, Vol. 31, No. 2, 244-248, 2008

ESTUDO DAS EMISSÕES DE FONTES MÓVEIS NA REGIÃO METROPOLITANA DE PORTO ALEGRE, RIO GRANDE DO SUL

Elba Calesso Teixeira*, Sabrina Feltes e Eduardo Rodrigo Ramos de Santana
Fundação Estadual de Proteção Ambiental Henrique Luis Roessler-RS, Rua Carlos Chagas, 55,
90030-020 Porto Alegre – RS, Brasil

2004
CO 195,740 (t/y)

ESTUDO DAS EMISSÕES VEICULARES NA REGIÃO METROPOLITANA DE PORTO ALEGRE

Leonardo Ribeiro P. Araújo[1,2], Eduardo Santana[1] (orient.) e Jonas da Costa Carvalho[2] (co-orient.)

2004
CO 160,015 (t/y)

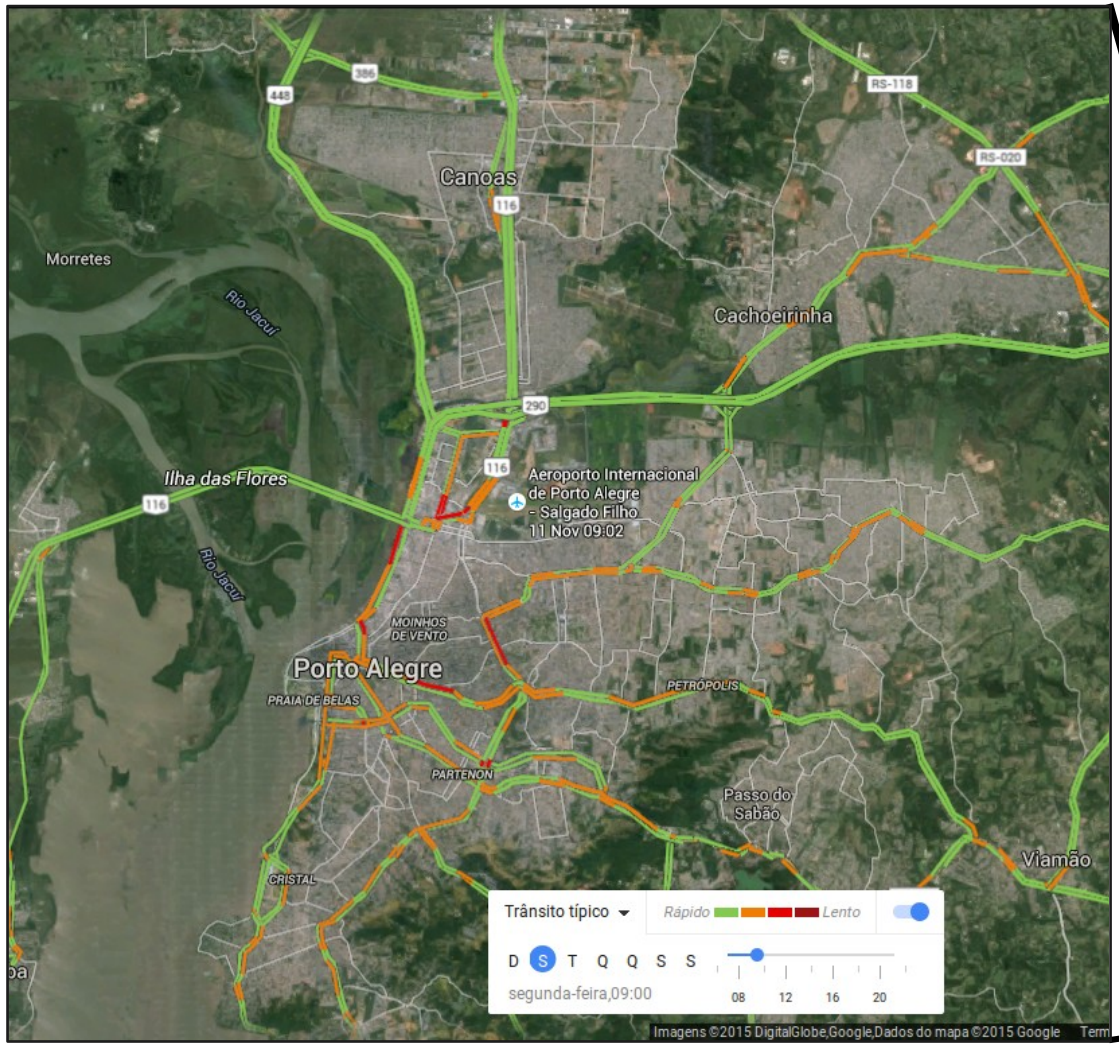
MODELAGEM DA POLUIÇÃO FOTOQUÍMICA NA REGIÃO METROPOLITANA DE PORTO ALEGRE.

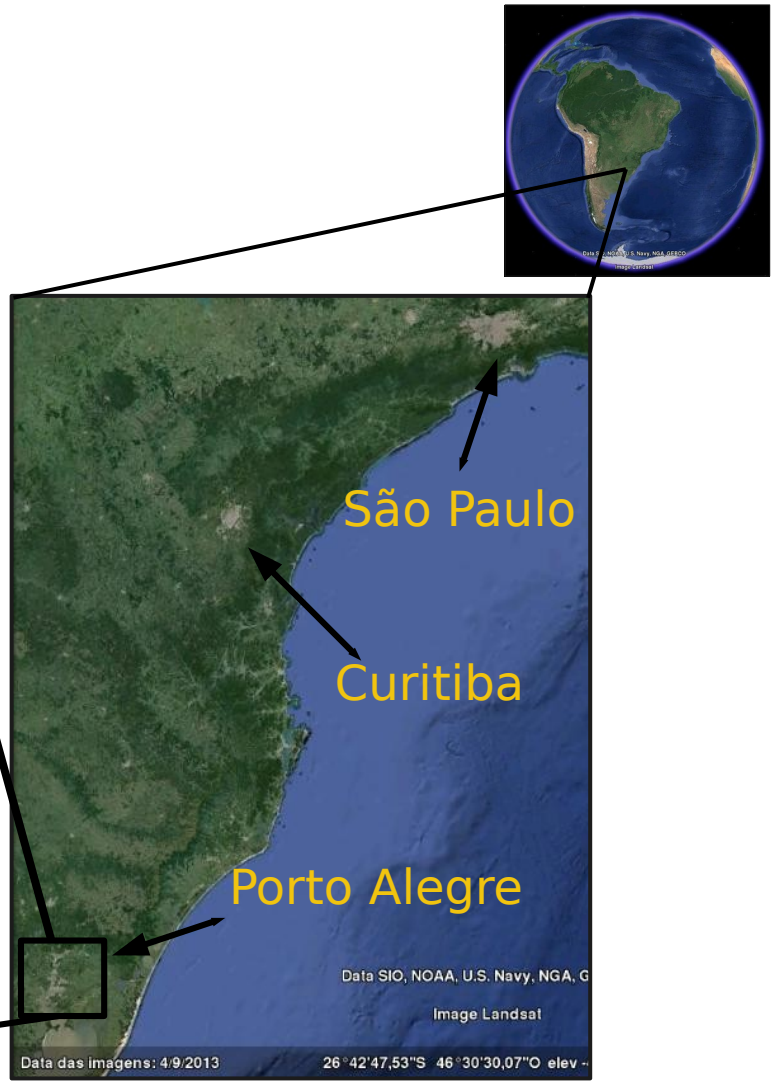
Gustavo Copstein Cuchiara[1]
Bruno Rodrigues[1]
Jonas da Costa Carvalho[1]

2009
CO 144,883 (t/y)

Porto Alegre City (2012)

- 2,7 milions hab
- 7th GDP
- 957.838 cars





Emissions inventories sources

Point

Industrial

Areal

Mobile

On Road

Off Road

Objectives

- **To develop a bottom-up (by street, hour and day) vehicular emissions inventory for MR POA**

- **Model emissions with WRF Chem**

Material and Methods

- **Road network from Open Street Map**
- **remIAG_0.3.3**
- **WRF Chem 3.6**
- **NCEP FNL. 2012/6/6-2012/6/9**
- **Fuel sales from ANP**
- **R 3.2.2 ggplot2, ggmap, maptools, ncdf, rgeos**

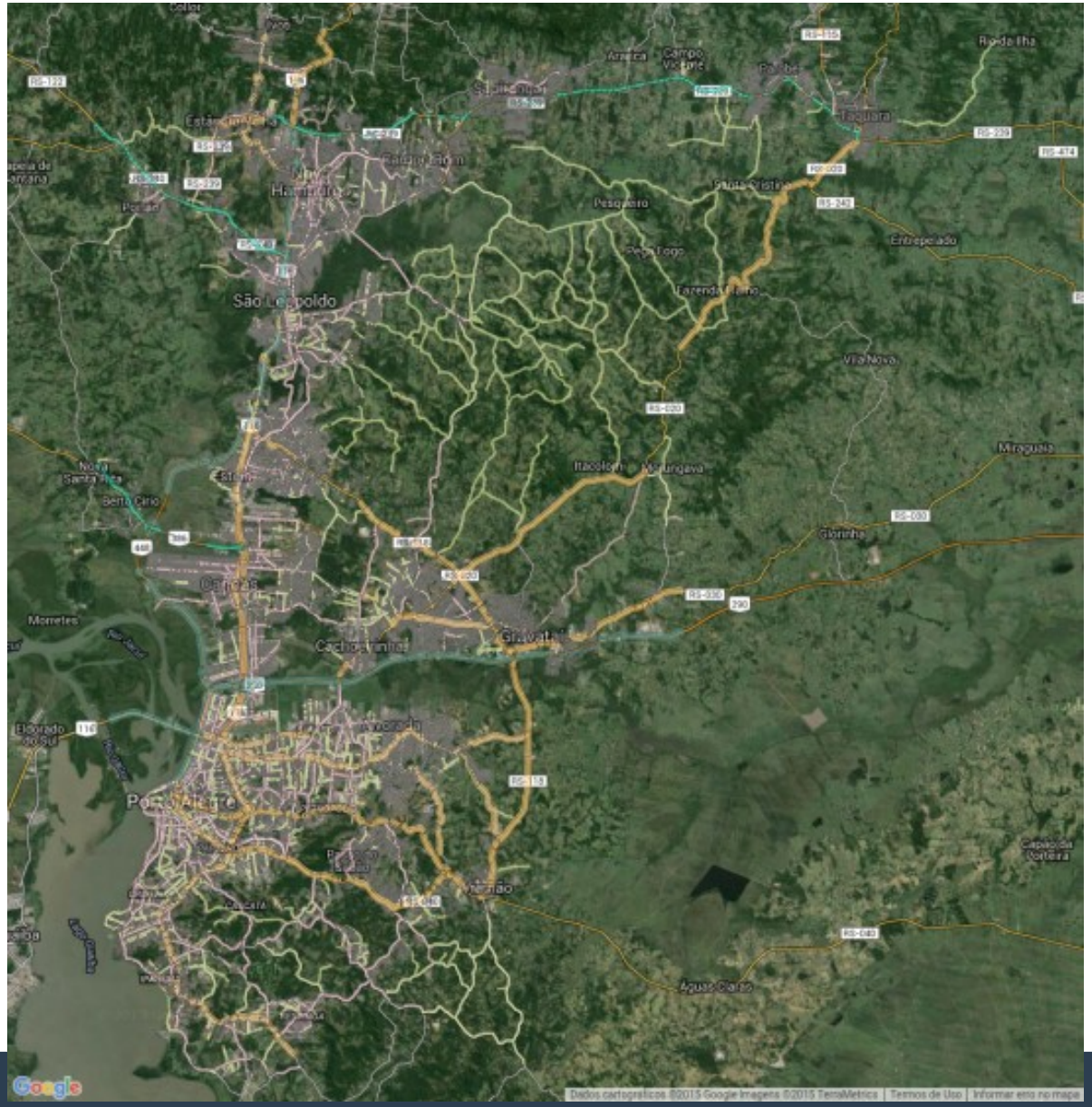
remIAG_POA

- **Traffic depends on type of street and distance from center**
- **Based on traffic count of São Paulo**

Street	PC	PC_10km	T	T_10km	B	M	M_10km
Motorway	3840	2569	169	225	66	834	743
Trunk	2980	1994	104	135	67	575	512
Primary	2313	1548	64	91	76	396	353
Secondary	1795	1201	43	67	87	273	243
Tertiary	1393	932	8	28		188	167
Residential	24	16	7	24		3	2
Average Age	14		18		5	6	

Passenger Cars and Light Commercial Vehicles, Trucks, Urban and Rented Bus and Motorcycles

Road Network



- poa
  - motorway
  - motorway_link
  - primary
  - primary_link
  - residential
  - road
  - secondary
  - secondary_link
  - tertiary
  - tertiary_link
  - trunk
  - trunk_link

remIAG: R Emissions inventory IAG

Which data do I have?

Local data:
Fleet Stats
EF/DF Cetesb

24h7d profiles

Traffic simulation Road Network

Traffic Counts Road Network

Road Network

A

Interpolation

Convert to EEA Format

Hourly-Link Emissions

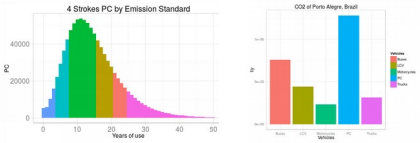
Tabplot

Emissions Street-based 24h 7d

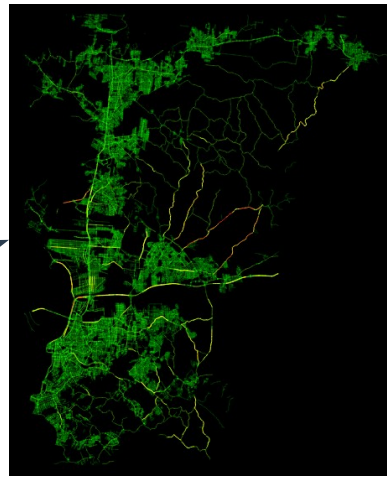
Export Shape

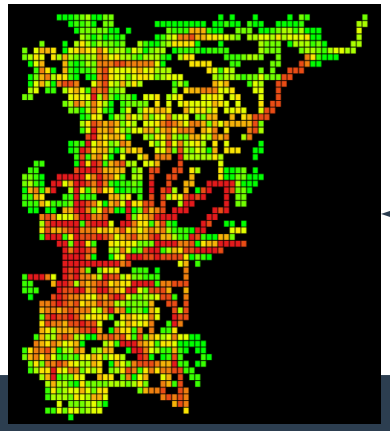
WrfChemMec

RADM2 SORGAM etc



NAME	TV	FUEL	SIZE	STANDARD	YEAR	CO_ty	NOx_ty
E_CO_A_G_L5_1000_ty	PC	Gasoline	1000	L5	2012	2.479308e+01	2.545504e+00
E_CO_A_FG_L5_1000_ty	PC	Flex Gasoline	1000	L5	2012	1.071360e+01	7.174095e-01
E_CO_A_FE_L5_1000_ty	PC	Flex Ethanol	1000	L5	2012	1.924003e+01	1.746737e+00
E_CO_A_G_L5_1000_2000_ty	PC	Gasoline	1000a2000	L5	2012	3.173059e+01	3.257778e+00
E_CO_A_FG_L5_1000_2000_ty	PC	Flex Gasoline	1000a2000	L5	2012	1.371144e+01	9.181525e-01
E_CO_A_FE_L5_1000_2000_ty	PC	Flex Ethanol	1000a2000	L5	2012	2.462371e+01	2.235503e+00
E_CO_A_G_L5_2000_ty	PC	Gasoline	2000	L5	2012	3.411891e-01	3.502987e-02



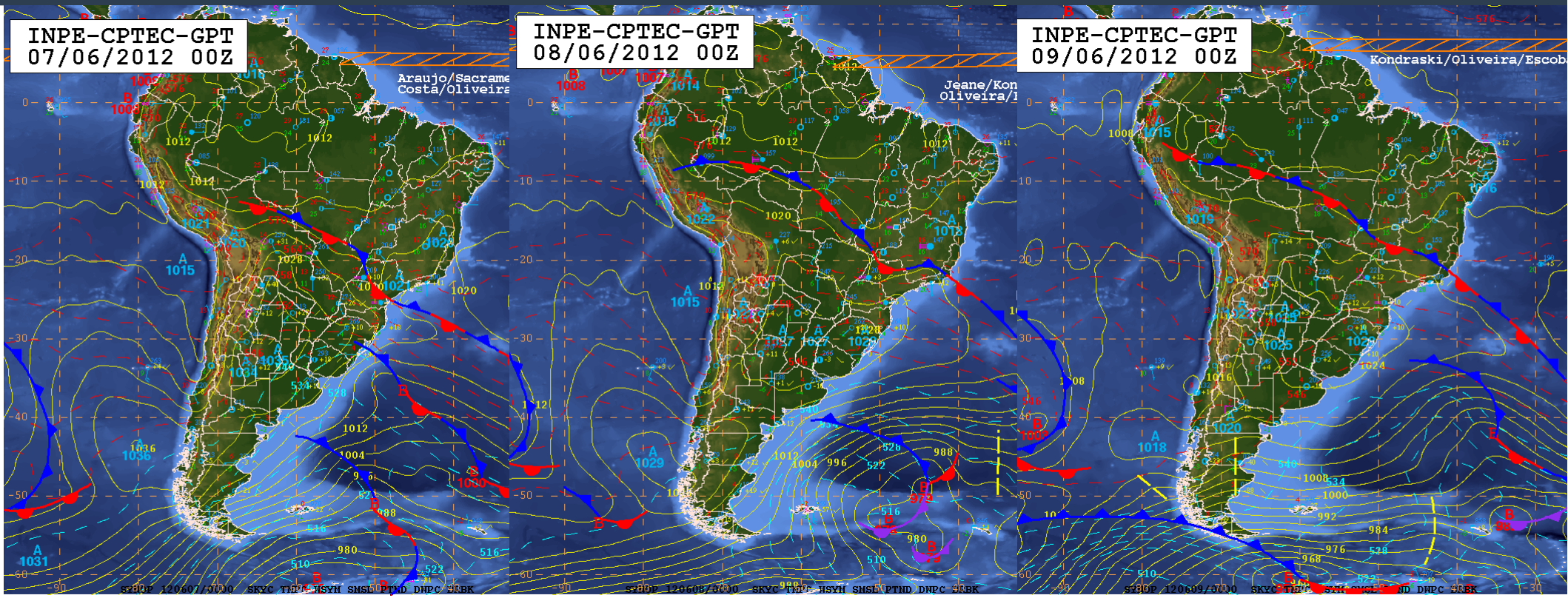


Model Setup

- 2012-06-06 00:00
  2012-06-09 00:00
- e_we = 60
- e_sn = 61
- ΔX = ΔY = 1000m
- 12 file emissions for monday
- 1 hr output

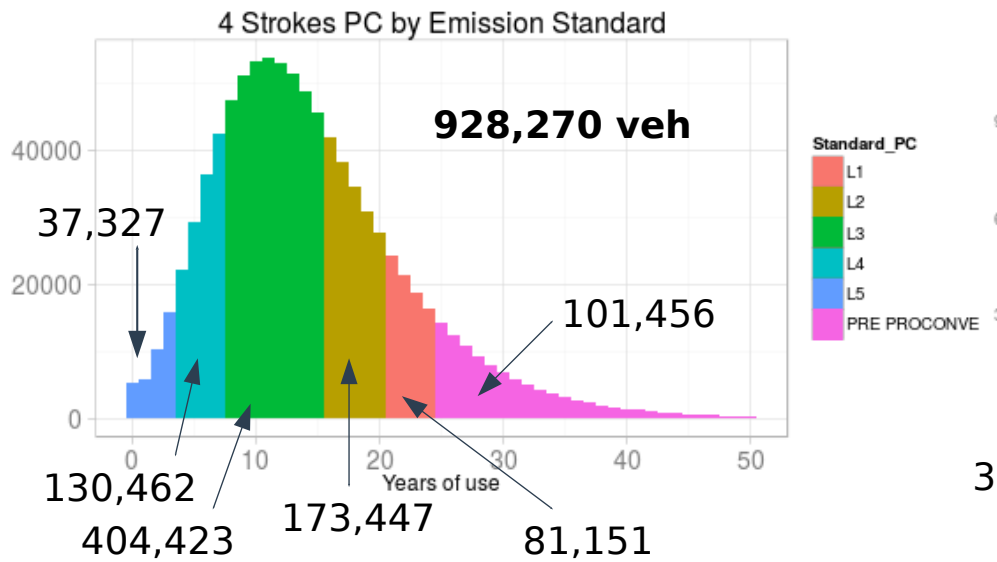
Atmospheric Process	Wrf-Chem option
Longwave radiation	RRTM
Shortwave radiation	Goddard
Surface layer	Monin-Obukhov
Land surface	Noah
Boundary layer	YSU
Cumulus clouds	Grell 3D
Cloud microphysics	Lin
Gas-phase chemistry	RADM2
Aerosol chemistry	MADE/SORGAM
Photolysis	Fast-J

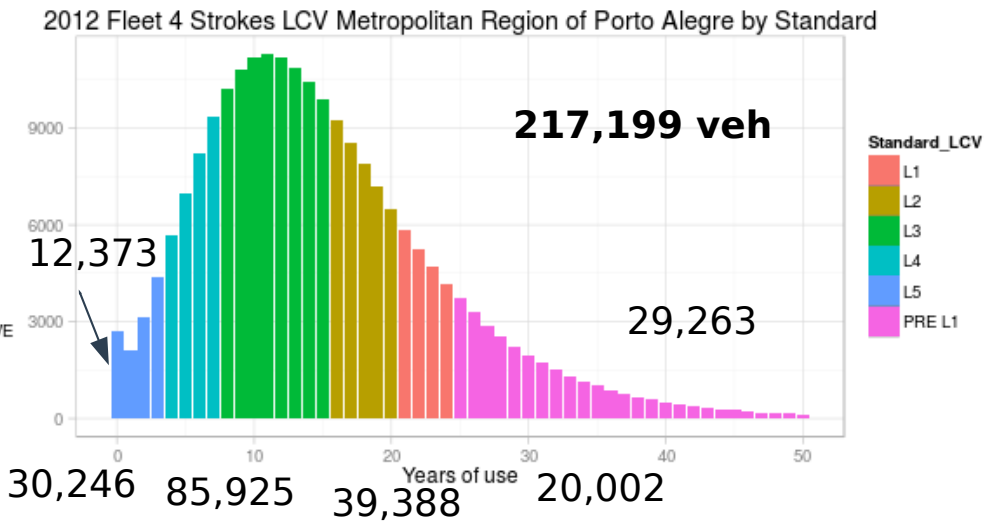
Synoptic conditions

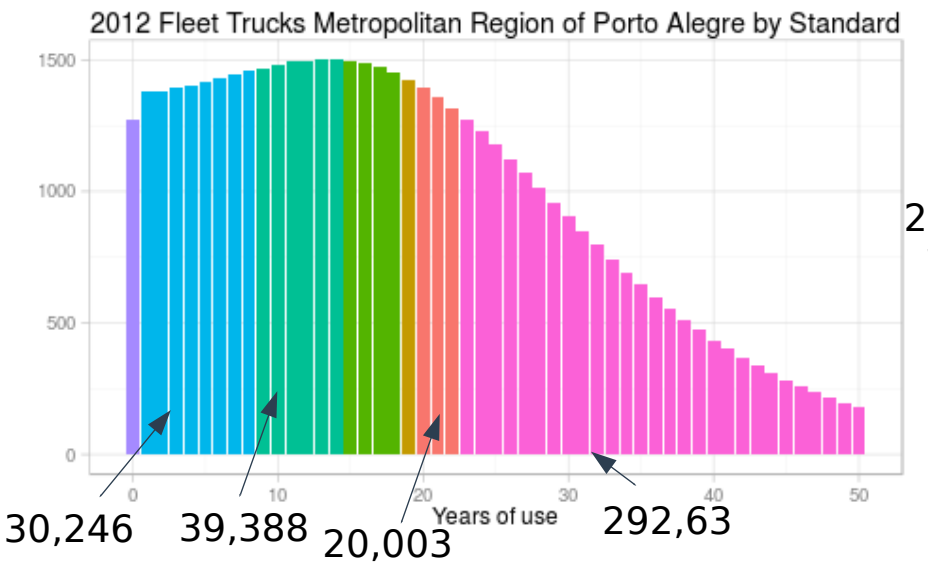


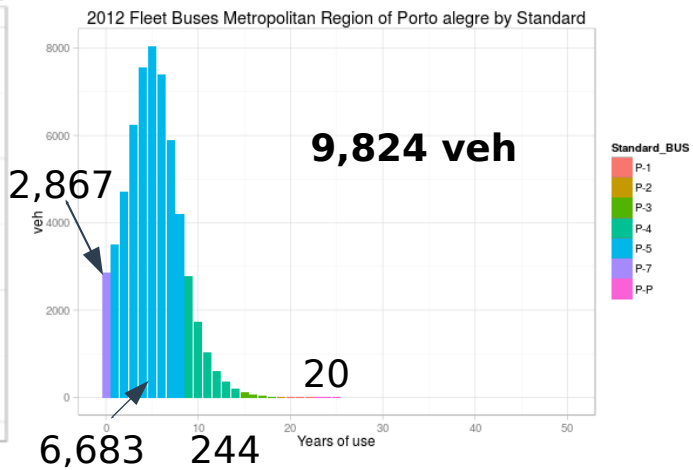
Day one front moved to the east of RS
Day two and three high in RS

remIAG_POA - Vehicles

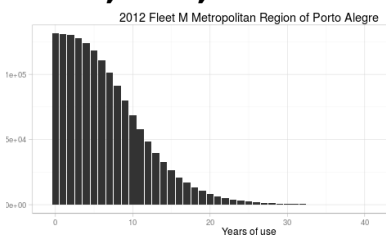








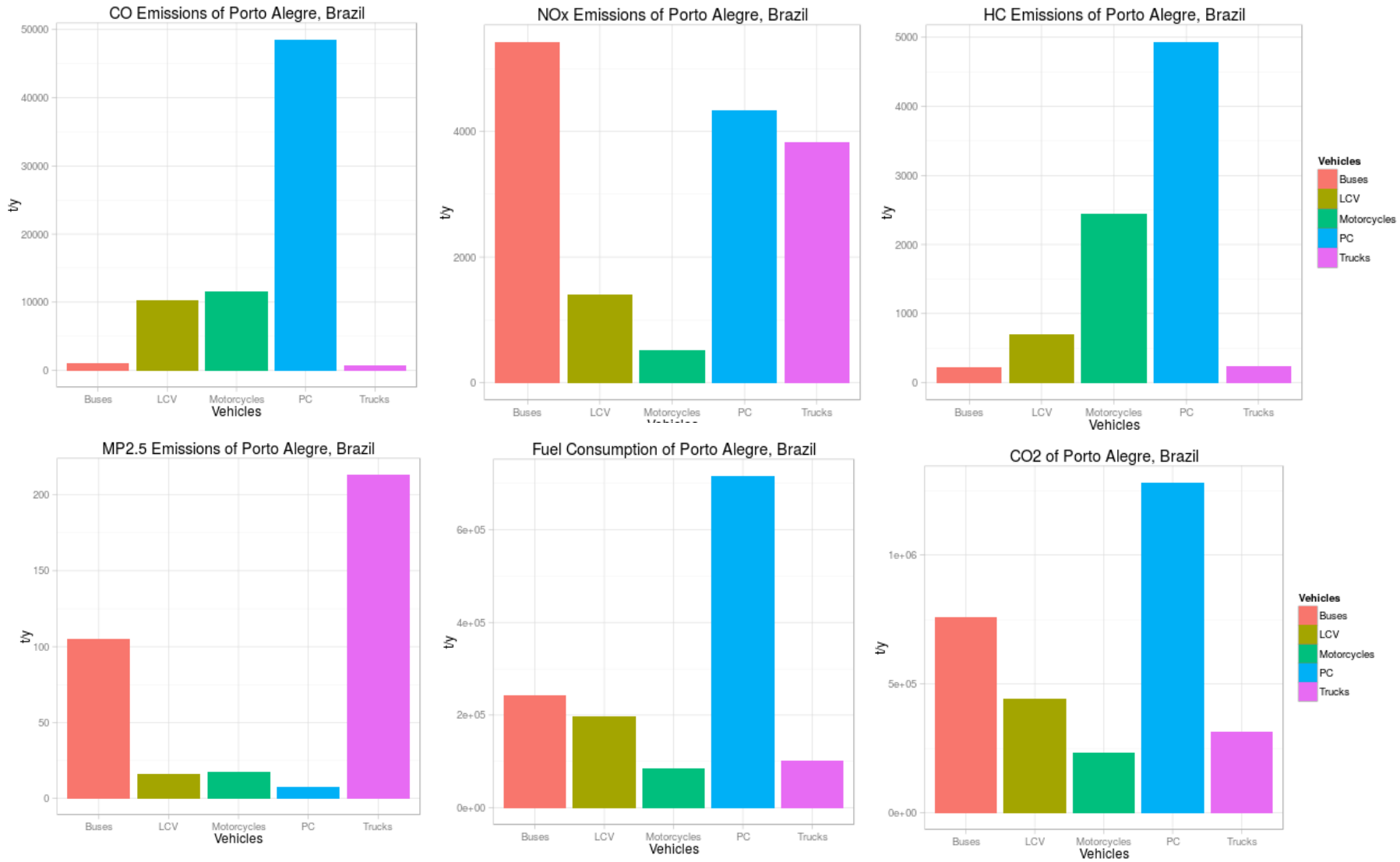
1,519,981 veh



remIAG_POA – Emissions (t/y)

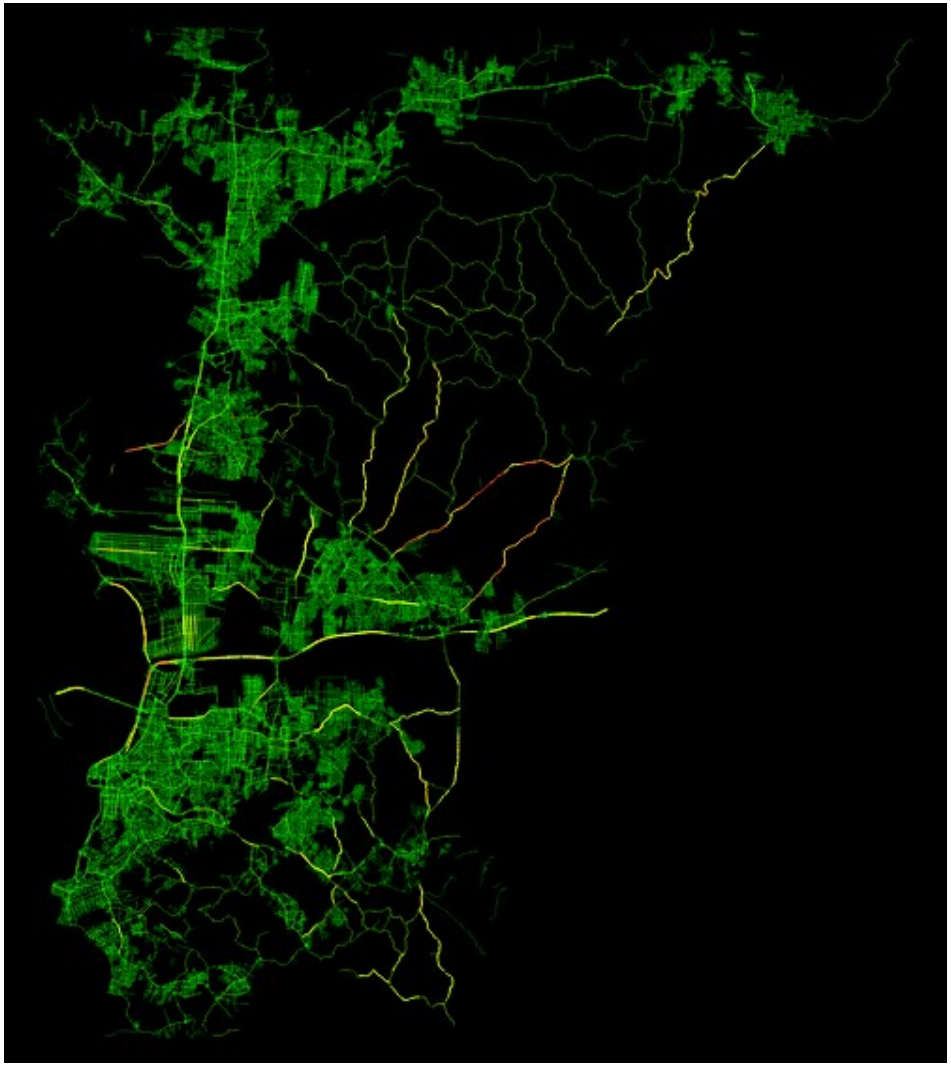
Vehicles	CO	NOx	HC	MP2_5	FC
PC	50342	4474	5096	7	742092
LCV	10316	1419	697	16	197271
Trucks	748	4048	261	226	106220
Buses	1031	5364	225	104	239120
Motorcycles	11562	514	2439	18	85767
Total	**73999**	**15819**	**8718**	**371**	**1370470**
Vehicles	**NO2**	**NO**	**HCNM**	**SO2**	**CO2**
PC	178	4297	4837	297	1327339
LCV	143	1276	588	79	447212
Trucks	461	3587	177	42	333454
Buses	724	4639	169	96	750662
Motorcycles	19	495	1937	34	233961
Total	**1525**	**14294**	**7708**	**548**	**3092626**

remIAG_POA - Emissions

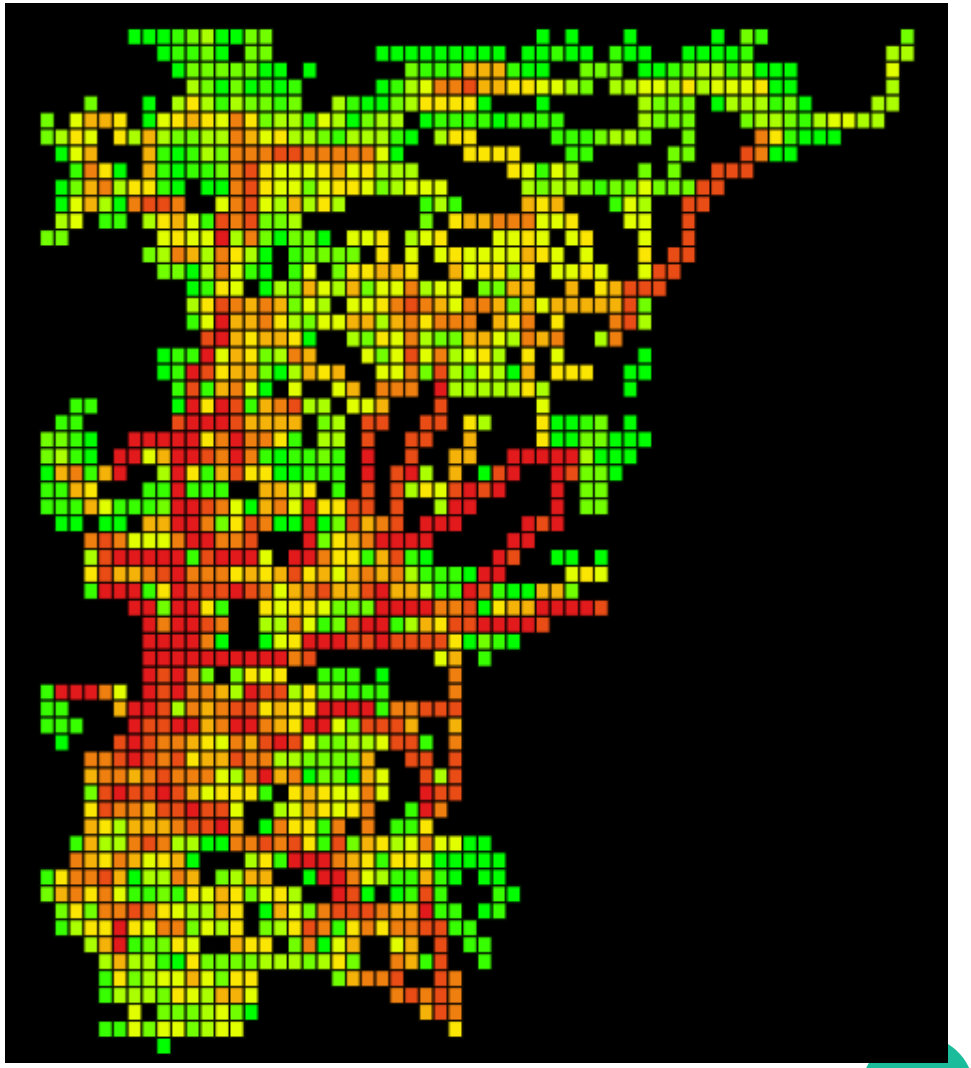


remIAG_POA - Emissions

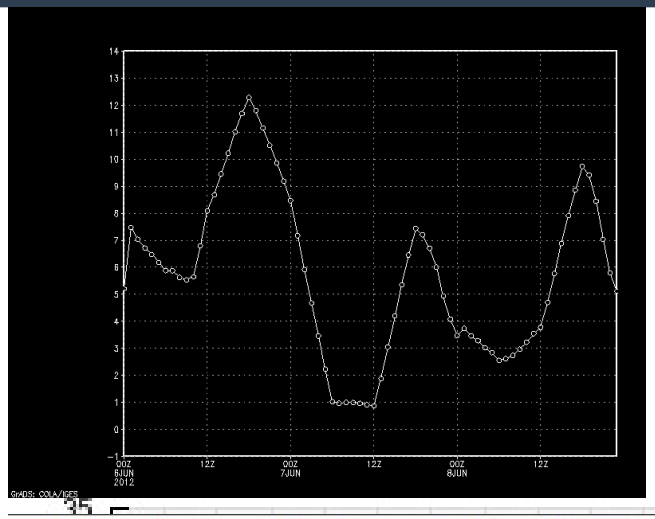
CO Monday 08:00 (g/km/h)

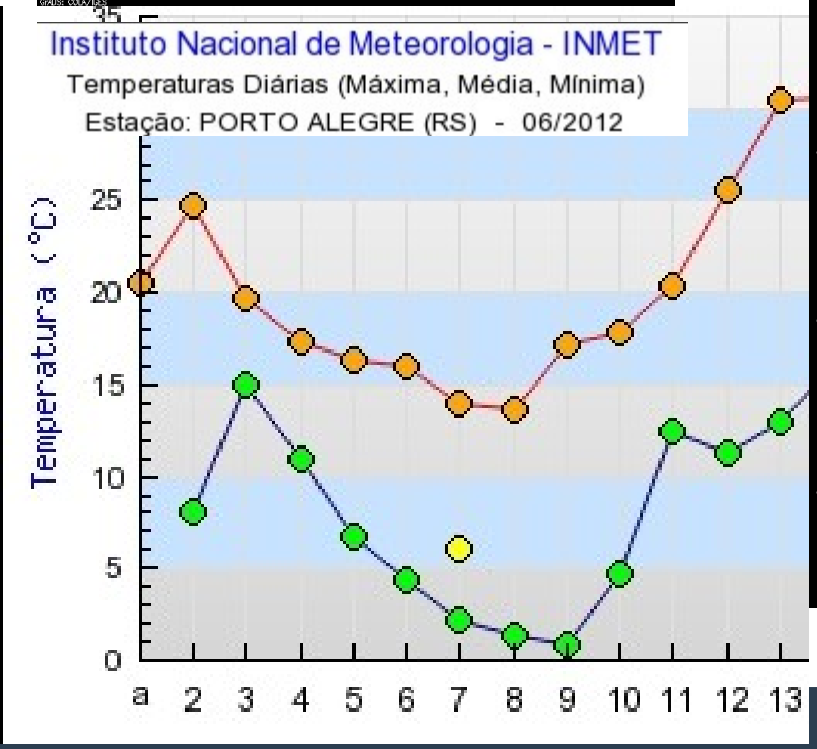


CO Monday 08:00 (mol/km^2/h)

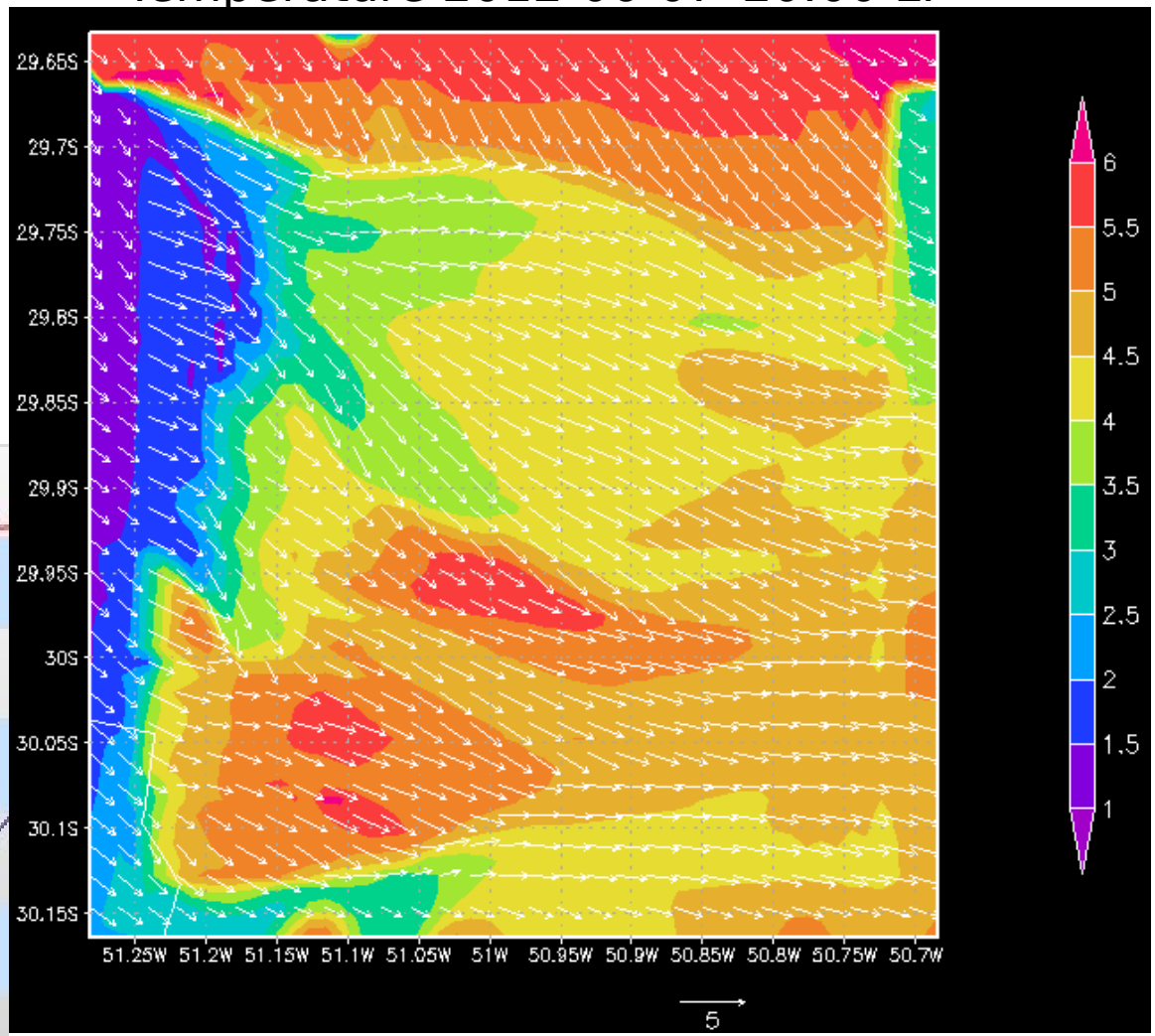


Temperature and Wind speed

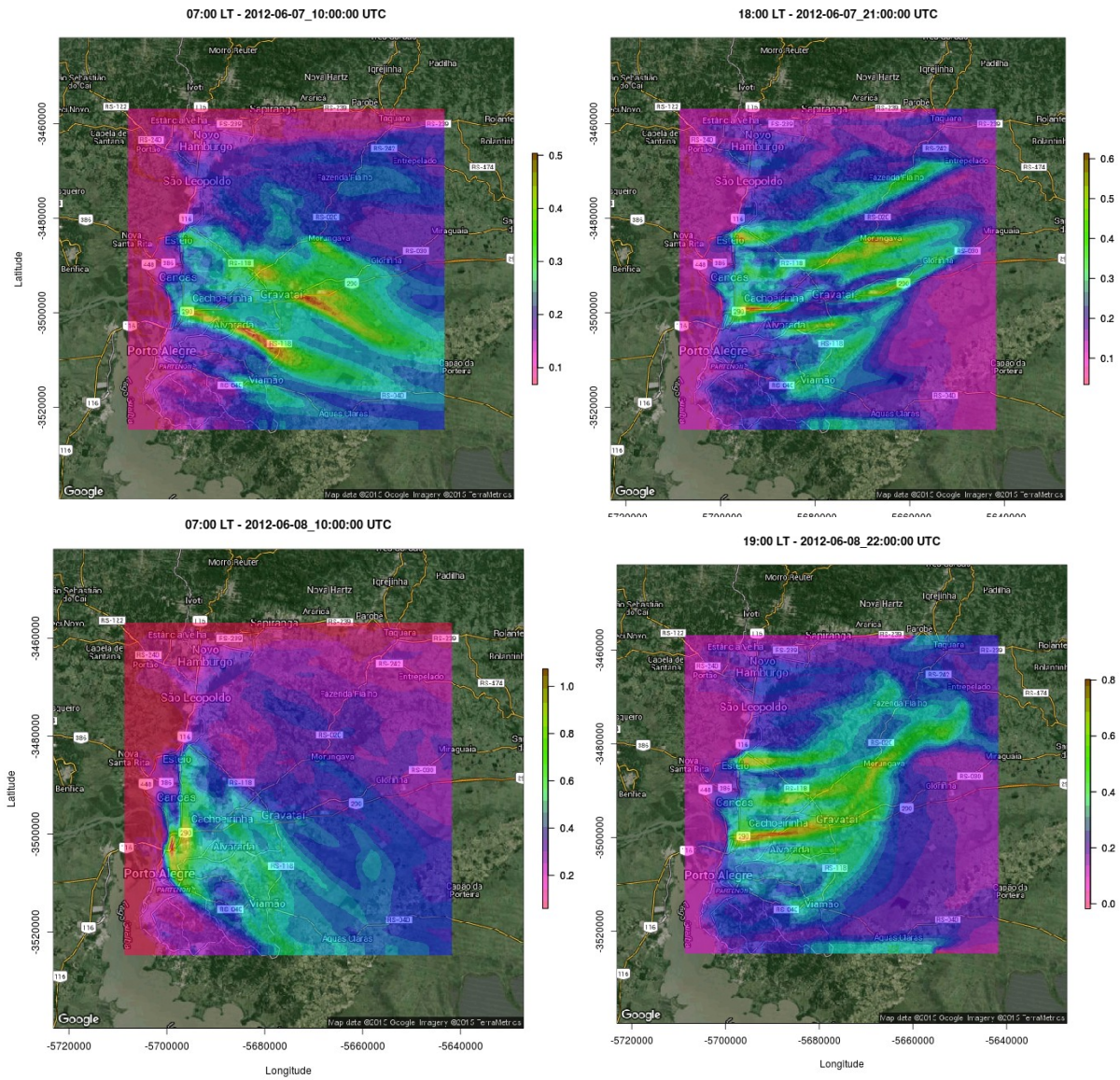




Temperature 2012-06-07 20:00 LT

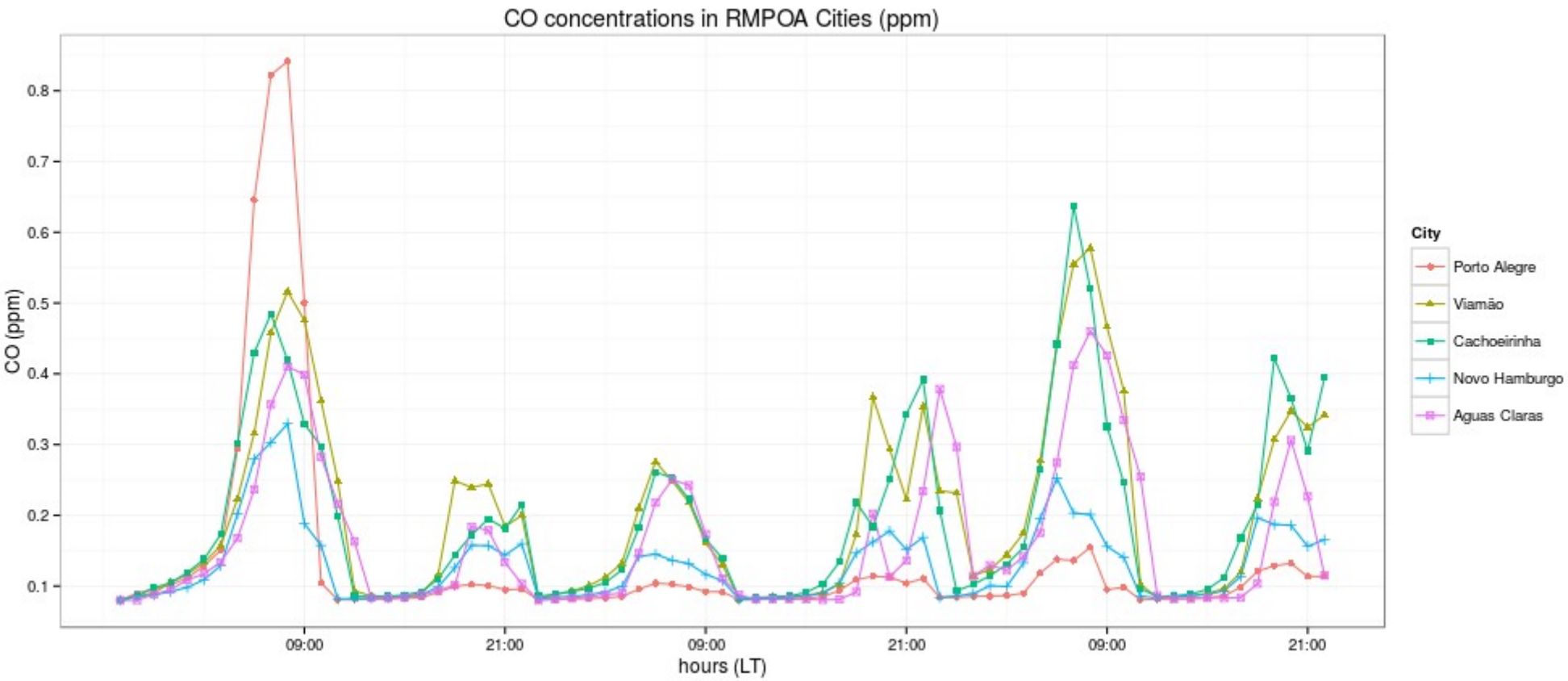


CO Concentrations – Days 2 and 3



Morning and peak hours

CO Concentrations – Cities



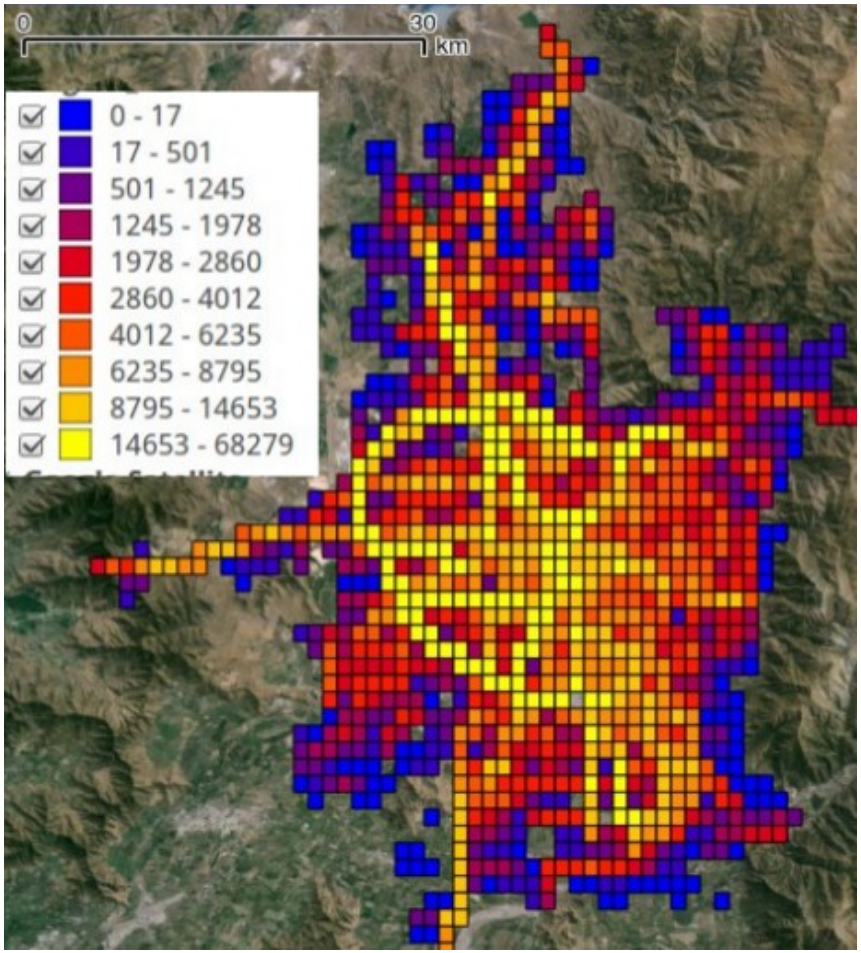
Discussion

Year	CO	NOx	HC	MP2.5	SO2	Studies
2012	73999	15819	8718	371	548	Ibarra (2015)
2004	195740	34110	78510	2350	1890	Teixeira (2008
2004	160015					Ribeiro (2008)
2009	144883	27979	11196	1051		Cuchiara (2009)

- **Vehicular emissions for 2012 looks smaller. Possible reasons:**
  - Technological improvements (PROCONVE)
  - Different methods, DF, EF, etc
  - Average age of fleet and lack of traffic counts.
  - Higher number of motorcycles is an example of difficulty of this approach
- **Air pollution simulation**
  - Simulation 1x1 degree to 1x1 km give plausible results, but it needs nesting
  - Difficulty to get air pollution data

Where does this was applied?

Santiago (Ibarra, 2015)



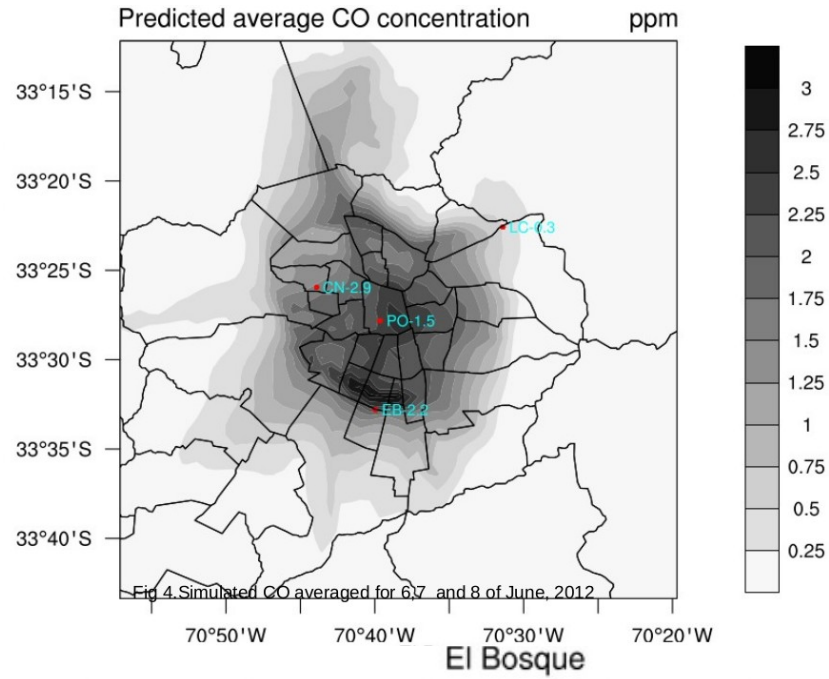
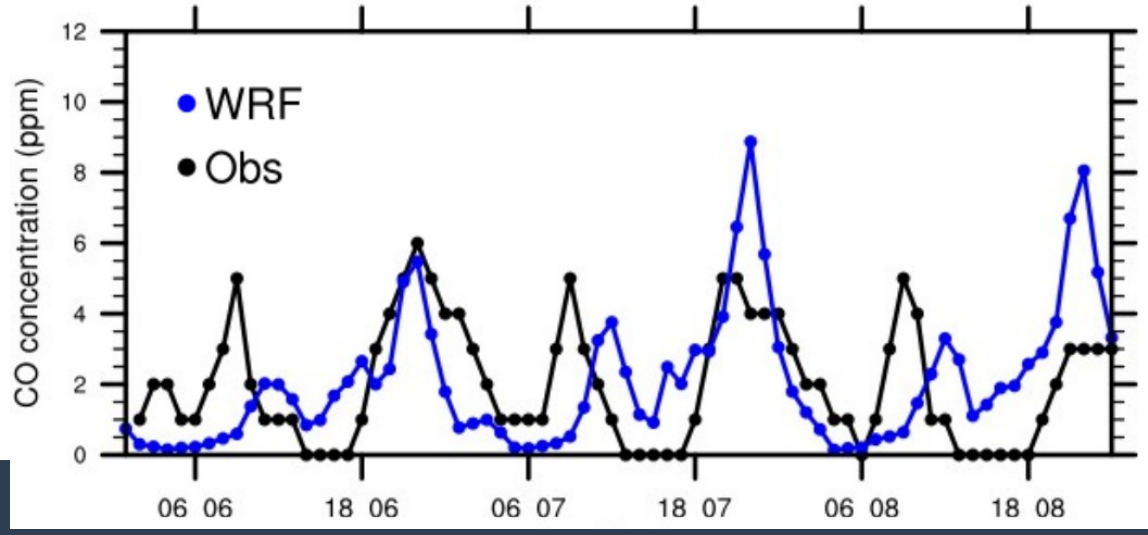
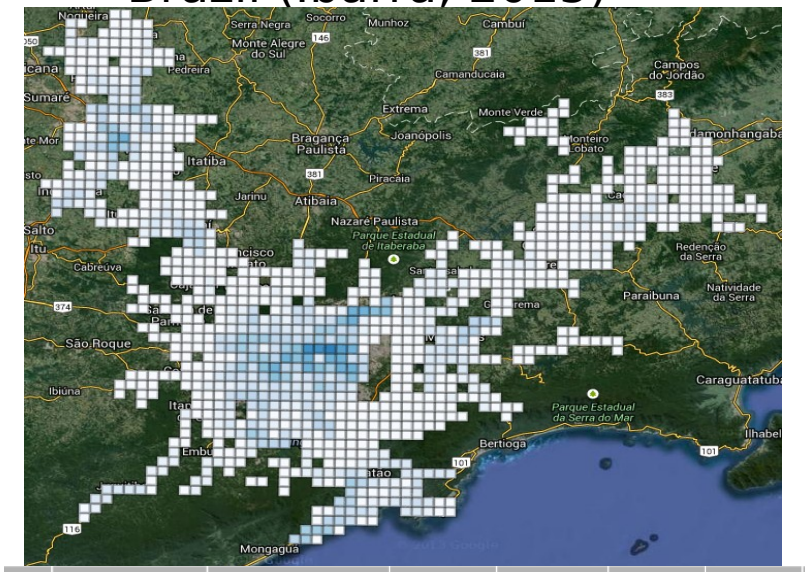


Fig 4.Simulated CO averaged for 6,7 and 8 of June, 2012

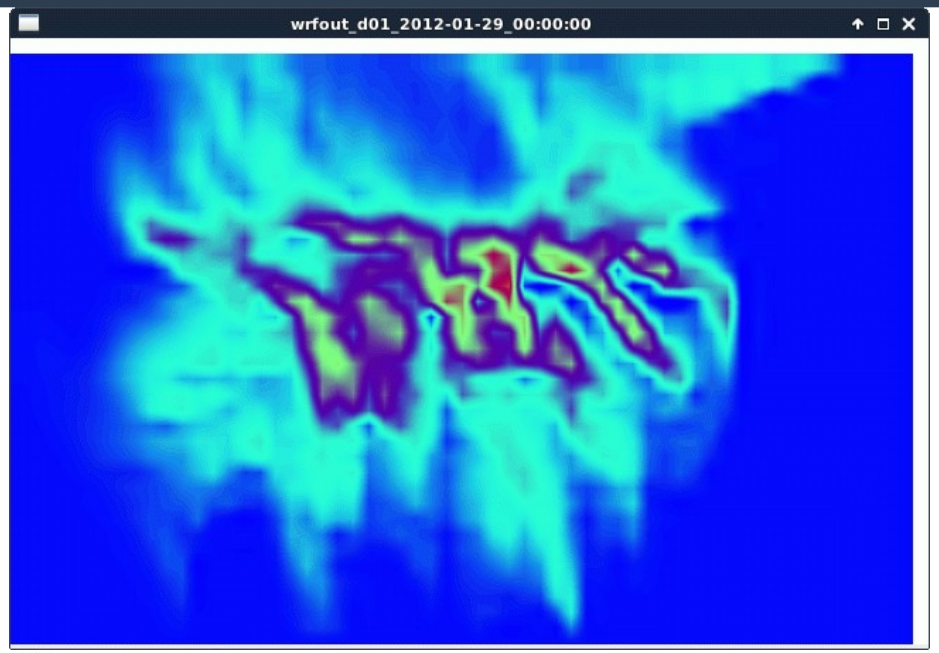


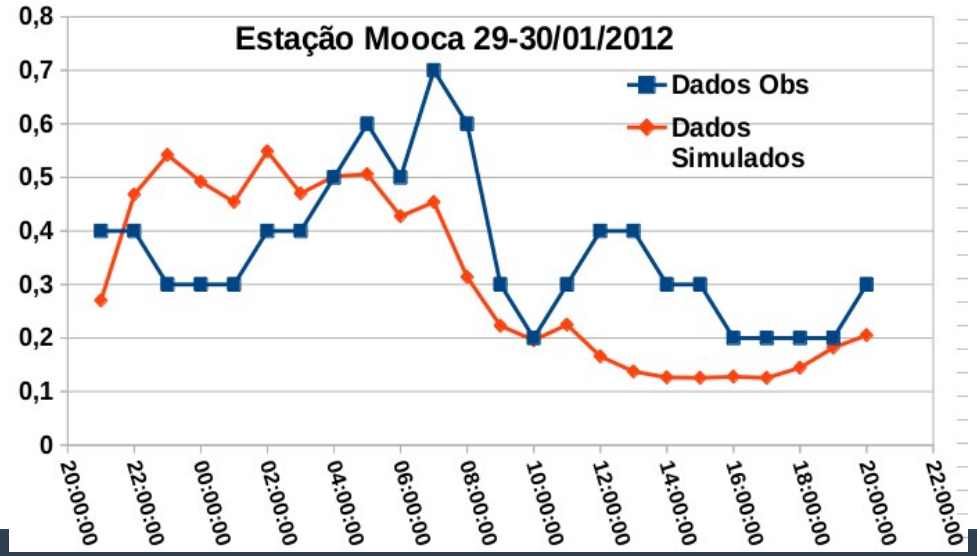
Where does this was applied?

Brazil (Ibarra, 2015)



ID	Countries	Main Citiy	MP2.5	CO	NOx	HC	Length
8	Brazil	Belem	112	30381	4237	2872	2359
9	Brazil	Brasilia	3797	303119	90092	34029	11696
10	Brazil	Campinas	1354	179127	38830	23850	10676
11	Brazil	Curitiba	723	141631	23987	16233	6833
12	Brazil	Fortaleza	559	1802	10009	622	5275
13	Brazil	Goiana	805	124970	23952	13487	9366
14	Brazil	Maceio	25	7801	1008	780	1213
15	Brazil	Manaus	139	25404	4201	2488	3013
16	Brazil	Porto Alegre	778	93098	20297	11320	7972
17	Brazil	Recife	224	47599	7624	5883	3772
18	Brazil	Rio de Janeiro	1529	151983	38014	20436	18260
19	Brazil	Salvador	397	60233	11927	7251	3438
20	Brazil	Santos	495	74561	15094	9044	3165
21	Brazil	Sao Luis	71	12671	2190	1330	1603
22	Brazil	SJDC	534	75876	15691	9682	3735
23	Brazil	Sao Paulo	1931	278446	55644	38142	26440
24	Brazil	Vitoria	276	48413	8327	5473	3658





Conclusions

- remIAG gives plausivel results but need to be adapted to every city with available data, adjusting equations.
- Emissions of POA follow an average city pattern.
- Results and model available in 2016 1x1 km.
- Southamerican results in raster and remIAG will be available in 2016.

References

- Cuchiara G., Rodrigues B., Costa-Carvalho J. Modelagem da poluição fotoquímica na região metropolitana de porto alegre. IV Encuentro Sul Meteorologia.
- Teixeira E., Feltes S., Ramos de Santana E. 2008. Estudo das emissões de fontes móveis na região metropolitana de porto alegre, rio grande do sul. Quim. Nova, Vol. 31, No. 2, 244-248, 2008
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- Ibarra S., Vara A., Ynoue R. 2015. Vehicular bottom-up emissions inventory and atmospheric simulation for 58 urban centers of South America. International conference on southern hemisphere meteorology and oceanography. Santiago, Chile, October 5-9.
- Boletim Sinotico CPETEC. http://tempo.cptec.inpe.br/bol_tecnico.shtml